Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Infographics and Data Visualization with PowerPoint

TGS Ref No: TGS-2026064178

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
3.0	26 January 2025	Previous approved lesson-plan release.	Tertiary Infotech Academy Pte Ltd
4.0	15 August 2026	Rebuilt from the approved v10 deck and current TSC. Added Microsoft infographic-maker and Copilot AI Designer workflows, eight progressive activities, editable mock CSV/Excel/PPT evidence, native PowerPoint chart practice, source traceability and current slide mappings. Detailed procedures remain in the Learner Guide and activity folders.	Tertiary Infotech Academy Pte Ltd
5.0	16 August 2026	Reframed the course around an AI-first, source-grounded infographic workflow. Added Microsoft 365 Copilot and Gemini Notebook generation routes, PowerPoint conversion and comparison, data-analysis and text-to-image prompts, synthetic source packs, AI-generated learner assets and human verification gates. Migrated the package to current course code TGS-2026064178.	Tertiary Infotech Academy Pte Ltd

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Course Information

Course Title	Infographics and Data Visualization with PowerPoint
WSQ Course Reference	TGS-2026064178
TSC Reference	Infographics and Data Visualisation (TOU-DAT-4004-1.1-1)
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN: 201200696W)
Duration	1 day · 9:30am–6:30pm · 8 counted hours including 1 assessment hour
Mode	Instructor-led classroom or synchronous virtual delivery with individual and small-group activities
Methods	Visual teaching, demonstration, AI-assisted source analysis, guided practice, data verification, comparative tool evaluation, peer coaching and reflection
Equipment	PowerPoint, Excel, internet access, projector, approved AI assistant, Microsoft 365 Copilot and Gemini Notebook access where available

Learning Outcomes

- LO1: Draw insights from infographics to make data-driven decisions.
- LO2: Leverage PowerPoint infographics for articulating data insights to stakeholders.
- LO3: Assess the effectiveness of PowerPoint infographics based on allocated resources in accordance with business goals.

Assessment

- Written Assessment (WA-SAQ) — nine open-ended questions, 30 minutes, open book.
- Practical Performance (PP) — three progressive tasks, 30 minutes, open book.
- Assessment block: 5:30pm–6:30pm.
- Grading: Competent / Not Yet Competent.
- A minimum of 75% attendance is required. Learners must be assessed as Competent and complete the required digital attendance and TRAQOM steps.

Daily Schedule

Time	Duration	Topic / Activity	Method	Slides
9:30–9:50	20 min	Welcome, digital	Briefing / visual facilitation	Slides 10–12

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		attendance, trainer profiles, outcomes and assessment briefing		
9:50–10:45	55 min	LU1 foundations and Activity 1 — Use AI to Clean Data and Extract Insights	Visual teaching + demonstration	Slides 17–31
10:45–10:55	10 min	Tea break	Break	—
10:55–11:50	55 min	Activity 2 — Prompt an Evidence-Led Infographic Brief; planning and wireframe review	Guided activity + debrief	Slide 39
11:50–1:00	70 min	LU2 foundations and Activity 3 — Build from Microsoft Infographic Maker	Visual teaching + demonstration	Slides 44–59
1:00–2:00	60 min	Lunch break	Lunch	—
2:00–2:50	50 min	Activity 4 — Generate Visual Assets and Refine Them in PowerPoint; complete Activity 3 outputs	Guided activity + debrief	Slides 59–67
2:50–3:00	10 min	Tea break	Break	—
3:00–4:00	60 min	Activity 5 — Use AI Analysis to Build a Native PowerPoint Data Story	Guided activity + debrief	Slide 75
4:00–4:45	45 min	Activity 6 — Compare Microsoft 365 Copilot and Gemini Notebook	Guided activity + debrief	Slide 83

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4:45–4:55	10 min	Tea break	Break	—
4:55–5:25	30 min	LU3 review, Activity 7 — Peer Coaching and Story Review, and Activity 8 — Budget and Effectiveness Decision	Visual teaching + demonstration	Slides 88–108
5:25–5:30	5 min	Course wrap-up, TRAQOM, digital attendance and assessment readiness	Review / readiness	Slides 120–122
5:30–6:00	30 min	Written Assessment (WA-SAQ) — nine open-ended questions, open book	Summative assessment	Slide 13
6:00–6:30	30 min	Practical Performance (PP) — three progressive tasks, open book	Summative assessment	Slide 120

Counted training and assessment time: 480 minutes (8.0 hours); lunch is excluded.

Topic-by-Topic Facilitation Notes

LU1 — Introduction to Infographics · Slide 17

Purpose · Types · Audience · Metrics · Planning · Insight extraction

- An infographic combines concise text, data, shapes, colour, charts and icons to reveal information and relationships.
- Chronology, timeline, process, comparison, statistical, mind-map, numerical and hierarchical formats answer different questions.
- An AI-first workflow gathers approved sources, cleans and analyses data, extracts candidate insights, then routes verified evidence into visual generation.
- Insight extraction distinguishes what the chart shows, what it means, why it matters and what action is justified.

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Activity 1 — Use AI to Clean Data and Extract Insights · Slide 31

Objective: use AI to organise and clean approved mock sources, then verify and convert candidate patterns into defensible business insights.

Method: Case study / guided workshop · Duration: 35 minutes · Output: A cleaned analysis workbook, AI prompt/source log, annotated insight slide and one-sentence recommendation.

Facilitation: teach the theory and visual examples first. Introduce the scenario from the deck; learners follow the detailed procedure in the Learner Guide and self-contained activity folder. Debrief against the acceptance check, not personal taste.

Activity 2 — Prompt an Evidence-Led Infographic Brief · Slide 39

Objective: turn verified insights into a structured AI brief and prompt that controls audience, message, evidence, form and output.

Method: Case study / guided workshop · Duration: 35 minutes · Output: A structured prompt brief, two AI wireframe directions and one approved PowerPoint wireframe.

Facilitation: teach the theory and visual examples first. Introduce the scenario from the deck; learners follow the detailed procedure in the Learner Guide and self-contained activity folder. Debrief against the acceptance check, not personal taste.

LU2 — Create Infographics in PowerPoint · Slide 44

Slide size · Grid · Shapes · Icons · Type · Colour · SmartArt · Charts · AI

- Visual hierarchy uses scale, position, contrast, spacing and alignment to guide the reader through one intended path.
- PowerPoint is the editable refinement layer for AI-generated structure, imagery, text, shapes, SmartArt, native charts and Excel-linked data.
- Microsoft 365 Copilot can generate an infographic from a prompt or file; Gemini Notebook can turn selected sources into an infographic or slide deck for PowerPoint refinement.
- Every AI-assisted output needs source verification, privacy review, brand control, accessibility checks and editable hand-off.

Activity 3 — Build from Microsoft Infographic Maker · Slide 59

Objective: adapt a Microsoft infographic template while preserving hierarchy, editability, brand and evidence.

Method: Case study / guided workshop · Duration: 40 minutes · Output: An editable portrait template-based infographic in PPTX, PDF and JPG.

Facilitation: teach the theory and visual examples first. Introduce the scenario from the deck; learners follow the detailed procedure in the Learner Guide and self-contained activity folder. Debrief against the acceptance check, not personal taste.

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Activity 4 — Generate Visual Assets and Refine Them in PowerPoint · Slide 67

Objective: use text-to-image prompting to generate a consistent asset set, then combine it with editable PowerPoint labels, shapes, connectors and SmartArt.

Method: Case study / guided workshop · Duration: 35 minutes · Output: A prompt log, consistent AI-generated asset set, editable five-stage infographic and reusable PowerPoint component slide.

Facilitation: teach the theory and visual examples first. Introduce the scenario from the deck; learners follow the detailed procedure in the Learner Guide and self-contained activity folder. Debrief against the acceptance check, not personal taste.

Activity 5 — Use AI Analysis to Build a Native PowerPoint Data Story · Slide 75

Objective: use AI to clean and analyse mock Excel data, verify calculations, then create a native editable chart and decision story in PowerPoint.

Method: Case study / guided workshop · Duration: 45 minutes · Output: An AI analysis log, verified formula workbook, native editable PowerPoint chart and annotated decision slide.

Facilitation: teach the theory and visual examples first. Introduce the scenario from the deck; learners follow the detailed procedure in the Learner Guide and self-contained activity folder. Debrief against the acceptance check, not personal taste.

Activity 6 — Compare Microsoft 365 Copilot and Gemini Notebook · Slide 83

Objective: generate the same source-grounded infographic brief in Microsoft 365 Copilot and Gemini Notebook, bring the Gemini Notebook deck into PowerPoint through PPTX download or a documented PDF rebuild route, and compare both routes before refinement.

Method: Case study / guided workshop · Duration: 45 minutes · Output: A Microsoft output, Gemini Notebook infographic and downloaded PPTX or PDF, comparison scorecard, reviewed final PowerPoint and prompt/source log.

Facilitation: teach the theory and visual examples first. Introduce the scenario from the deck; learners follow the detailed procedure in the Learner Guide and self-contained activity folder. Debrief against the acceptance check, not personal taste.

LU3 — Review Infographics · Slide 88

Story · Context · Simplicity · Coaching · Budget · Effectiveness · Improvement

- Effective review separates message clarity, data integrity, visual hierarchy, accessibility, audience fit and business usefulness.
- Coaching uses specific observations, questions and criteria rather than taste-based comments such as 'make it pop'.
- Tool investment is evaluated through capability, licence, learning, production, governance, collaboration and maintenance costs.

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- Improvement is evidence-led: gather feedback, prioritise changes, revise the controlled master and retest the intended outcome.

Activity 7 — Peer Coaching and Story Review · Slide 99

Objective: give specific criteria-based feedback and revise an infographic to improve clarity, story and accessibility.

Method: Role play · Duration: 30 minutes · Output: A completed peer-review log, revised infographic and short reflection.

Facilitation: teach the theory and visual examples first. Introduce the scenario from the deck; learners follow the detailed procedure in the Learner Guide and self-contained activity folder. Debrief against the acceptance check, not personal taste.

Activity 8 — Budget and Effectiveness Decision · Slide 108

Objective: compare infographic-tool investment options and recommend a resource plan using effectiveness and governance criteria.

Method: Case study / guided workshop · Duration: 30 minutes · Output: A weighted tool-investment scorecard and one-page resource recommendation.

Facilitation: teach the theory and visual examples first. Introduce the scenario from the deck; learners follow the detailed procedure in the Learner Guide and self-contained activity folder. Debrief against the acceptance check, not personal taste.

Resources and Trainer Preparation

- Open the final PPTX and backup PDF; verify links, transitions, presenter display and current version.
- Confirm every Activity-01 to Activity-08 folder contains instructions, checklist, PROMPTS.md, CSV, XLSX and PPTX assets.
- Verify workbook formulas, AI Prompt Lab sheets, Gemini Notebook comparison scorecard and native chart behaviour; retain all synthetic-data source notes.
- Confirm which Microsoft 365 Copilot and Gemini Notebook features are available for approved learner accounts; provide the supplied editable and offline source-pack fallbacks.
- Inspect Activity 04 generated assets and provenance manifest; remind learners that generated imagery is supporting material, never measured evidence.
- Prepare SSG digital-attendance and TRAQOM access. Keep assessment answer keys trainer-only and out of GitHub/LMS learner links.

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